From Land to Liquidity: Land Acquisition Compensation and Elite Mobility in Taiwan's Land Reform

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Motivation

- Taiwan's land reform has long been viewed as the root of the "Taiwan Miracle." However, existing studies find little improvement in agricultural output through property rights and only marginal effects through relaxation on restrictions of crop choice.
- Its effects are better understood through the transformation of social structure and the resulting political and economic differentiation.
- Land-to-the-Tiller converted land wealth into liquid assets, facilitating the optimal allocation of people across places and occupations. Different compensation rules for dry and wet land provides exogenous variations for causal interpretations.

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Three stages of postwar land reform

Year	Name	Land or contractual relationship	Administrative action
1949	375 Rent Reduction	Existing Tenant contracts	Rent ceiling, tenure protection, and contract registration
1951	Sale of Public Land	Government-held cultivated land already under tenancy	Sale to incumbent cultivators with installment payment
1953	Land-to-the-Tiller	Privately owned rented land above legal retention limits	Government acquisition and resale to incumbent tenants

Literature: tenancy and agricultural output

Cheung (1969) and Liu (2022)

- Cheung's share-tenancy analysis treats the 37.5% rent rule as a share of realized harvest.
- Liu shows that the ceiling used standard yield. Rent was fixed with respect to realized output.
- The usual sharecropping incentive argument therefore does not describe this rule well.

Kim and Wang (2025)

- They test the direct effect of reform on agricultural output and find little robust gain from the private-land transfer.
- Colonial extension and farmer training had already spread modern techniques. Another yield jump may have been limited.
- Freedom to change crops on former public land may offer a sharper output margin.

Literature: land as a financial and social asset

Koo's recollections

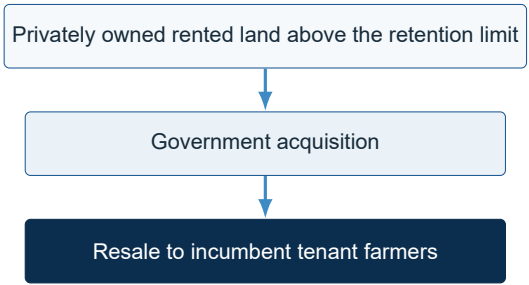
- The account collected in *Jin Han Mei Xiang* emphasizes the release of land wealth into more liquid assets.
- Koo participated closely in reform while standing outside government. His observations provide a distinct participant perspective.

Qu and He

- Qu traces landlord compensation into enterprise shares and industrial finance.
- He reads reform as a change from village-based relations to modern individual property rights and state governance.

This paper studies population reallocation, long-run employment structure, and local electoral outcomes.

The 1953 Land-to-the-Tiller transaction



Compensation to former landlords

- 70% of land value in commodity-denominated land bonds
- 30% in shares of state-owned enterprises
- Bonds paid 4% annual interest and were amortized over ten years
- Outside Penghu, repayment was scheduled twice per year

Statutory text: bond denomination & repayment asset

本債券券面實物，分稻穀與甘薯二類：一、稻穀債券，定額十二億六千萬公斤，用以補償徵收田地目部份所包括之兩季田、單季田、看天田、輪作田（包括特殊田）地價。二、甘薯債券，定額四億四千萬公斤，用以補償徵收畑地目部份地價。以上二類債券，由臺灣省政府於民國四十二年度內按照債券面額十足發行。

本債券到期本息，依左列規定償付之：一、稻穀債券，償付稻穀。但用以補償輪作田（包括特殊田）地價所搭發之稻穀債券，按該年期當地稻穀時價折付現金。二、甘藷債券，按該年期之甘藷時價折付現金。前項現金本息，由臺灣土地銀行當地行處經付，其稻穀本息，得轉託糧食局指定之當地倉庫經付，其經付稻穀標準，依臺灣省田賦徵實實收規則之規定。

Results in brief

- **Population:** -169.7 residents per 10 pp of township dryland share in the baseline diff-in-diff regression.
- **Migration:** $+3.842$ pp in residence outside the birth township per 10 pp of tenant dryland share. Migrants hold more financial assets and have higher economic status and human capital.
- **Employment:** -0.107 pp in financial and business services and -0.556 pp in next-year agricultural-to-nonagricultural transitions per 10 pp of tenant dryland share.
- **Elections:** $+0.050$ candidates, -0.672 pp in the winner's vote share and 2.300 pp in the probability of a KMT victory, per 10 pp of tenant dryland share.
- **Public land:** $+2.353$ pp in agricultural employment in pooled censuses per 10 pp of leased public-land share.

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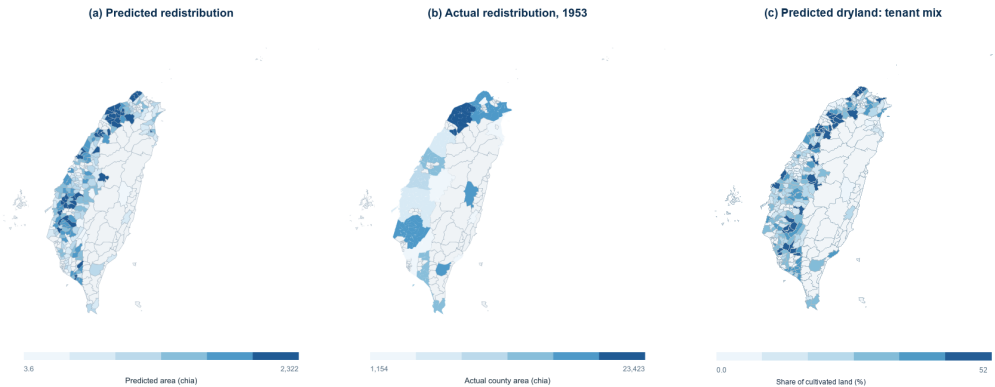
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Data sources and years

Source	Years
Colonial censuses and population statistics	1915–1935
County and municipal statistical yearbooks, statistical abstracts, household-registration records, and local gazetteers	1947–1954
《臺灣省戶籍統計要覽：民國三十五年至四十七年戶籍統計年報》	1955–1958
《耕地所有權與經營狀況調查報告書》	1950
Digitized research appendix by Chiang, Fan, and Hsu	1950, 1953
《農復會工作報告》第五期	1953
Population and housing censuses	1966, 1980, 1990, 2010, 2020
Taiwan Social Change Survey	2012, 2017, 2022
Manpower Utilization Survey	1978–2000
Internal Migration Survey	1980–1989, 1992
Central Election Commission	1998–2022

Predicted and actual redistribution across space



Research outcomes and land measures

Outcomes

- Registered township population
- Residence outside the birth township
- Employment composition and annual industry transitions
- Township mayoral election outcomes

Land measures from the 1950 survey

$$DryShare_i = \frac{\text{dryland}}{\text{all cultivated land}}$$

$$TenantDry_i = \frac{\text{tenant-operated dryland}}{\text{tenant-operated cultivated land}}$$

$$PublicShare_i = \frac{\text{leased public cultivated land}}{\text{all cultivated land}}$$

Predicted private redistribution is constructed from the owner-size distribution.

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Baseline continuous-treatment difference-in-differences

$$Population_{it} = \alpha_i + \lambda_t + \beta\big(DryShare_i \times Post_t\big) + \varepsilon_{it}$$

- For Land-to-the-Tiller, $Post_t = 1$ from 1952, when large-scale public communication began.
- Event studies omit 1951, the last year before the first treated year.
- Township and year fixed effects absorb local constants and Taiwan-wide shocks.
- Standard errors are clustered by the 237 treatment units.

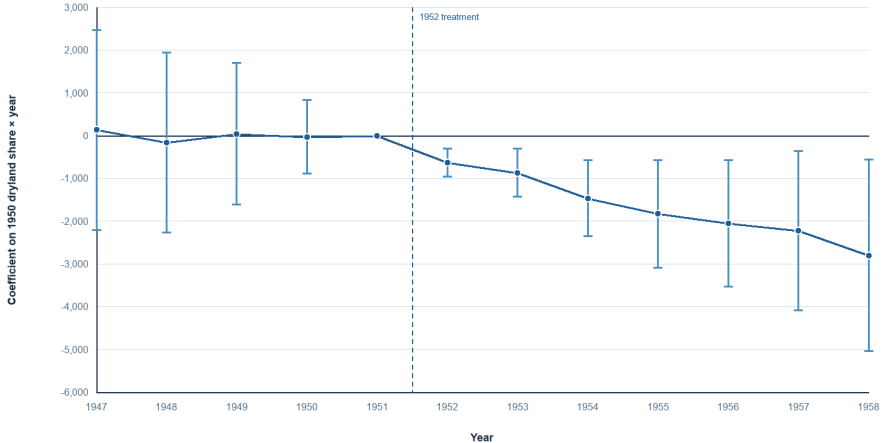
Year	1947	1948	1949	1950	1951	1952	1953	1954	1955	1956	1957	1958
Units	148	149	154	205	237	237	227	237	237	237	237	237

Baseline estimates and sample completeness

	All available (1)	95% coverage (2)	85% coverage (3)	Complete years (4)
$DryShare_i \times \mathbf{1}[t \geq 1952]$	-1697.273^{*}	-1702.579^{***}	-1692.290^{***}	-1836.304^{***}
	(867.396)	(622.278)	(650.008)	(685.846)
p-value	0.050	0.006	0.009	0.007
Township FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
Observations	2,542	1,886	2,091	1,659
Treatment units	237	237	237	237
Sample years	1947–1958	1951–1958	1950–1958	7 complete years

Dependent variable: population count. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Event-study estimates and pre-trend assessment



Error bars are unit-clustered 95% confidence intervals. Pre-treatment joint test for 1947–1950: $\chi^2(4) = 5.282$, $p = 0.260$. Post-treatment joint test for 1952–1958: $p < 0.001$.

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Construction of predicted redistribution

1. Approximating the land ownership distribution (lognormal) with MLE method, using data from the 1950 township land survey.
2. Calculate land above the retention threshold $h = 3$ chia.
3. Assume that the proportion of dry land being redistributed is equal to the proportion of dry land cultivated by the tenant farmers.

The 1951 sale of public land



Institutional distinction

- The 1951 program covered government-held cultivated land.
- Incumbent cultivators purchased the land in installments.
- Its timing predates the 1953 acquisition of privately owned rented land.

Empirical proxy

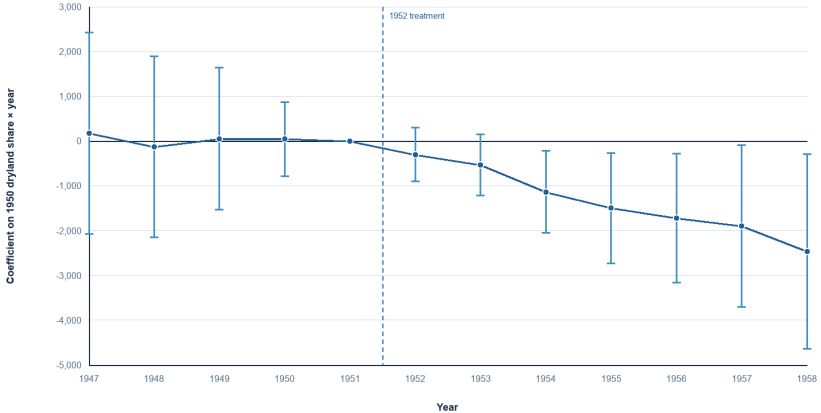
Leased public cultivated land in the 1950 survey, measured in chia and transformed using a standardized inverse hyperbolic sine.

Population effects: alternative dryland measures

	Township dryland ($DryShare$)			Tenant dryland ($TenantDry$)		
	Baseline	+ Public land	+ Both	Baseline	+ Public land	+ Both
	(1)	(2)	(3)	(4)	(5)	(6)
$DryShare_i \times Post_t$	-1697.273^{*} (867.396)	-1579.865^{*} (835.997)	-1395.399^{*} (803.064)	–	–	–
$TenantDry_i \times Post_t$	–	–	–	-3152.556^{***} (739.833)	-3378.528^{***} (668.775)	-3166.791^{***} (685.418)
$PublicLease_i \times Post_t$	–	542.464^{***} (169.941)	586.178^{***} (159.539)	–	603.627^{***} (163.237)	637.803^{***} (154.541)
$TotalRedist_i \times Post_t$	–	–	2766.786^{*} (1551.986)	–	–	2445.860 (1537.348)
Township FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Observations	2,542	2,542	2,542	2,542	2,542	2,542
Townships	237	237	237	237	237	237

Outcome: population count. $Post_t = 1$ from 1952. Public land: standardized asinh of 1950 rented public land. Redistribution: predicted area / cultivated area. Township-clustered SE in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Event study with both reform exposures controlled



Pre-treatment joint test $\chi^2(4) = 4.885$, $p = 0.299$

Post-treatment joint test $\chi^2(7) = 19.754$, $p = 0.006$

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Migration and migrants' characteristics

A. Dryland exposure and residence outside the birth township

	Tenant dryland	Township dryland
	3.842^{**}	5.336^{***}
Land share (per 10 pp)	(1.487)	(1.633)
Historical controls	Six	Six
County, year, and birth-year FE	Yes	Yes
Observations	692	692

B. Migrants compared with residents of their birth township

	College attendance or above (pp)	Professional/technical occupation (pp)	Household income rank (percentile)	Financial products held/invested in (pp)
	10.52^{**}	8.56^{**}	13.78^{**}	14.38^{**}
Migrant	(4.37)	(3.11)	(6.02)	(5.51)
Observations	690	664	426	532

Born before 1953. Migrant: current township differs from birth township. Survey weights; birth-county clustered SE in parentheses. Panel A includes crop output and double-cropping controls. Panel B includes year, birth-year, and birth-township FE. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

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Specification for long-run outcomes

$$y_{uit} = \alpha_{c(i)} + \lambda_t + \beta Z_i + X_i'\gamma + \varepsilon_{uit}$$

- Z_i: tenant dryland share, township dryland share, or leased public-land share. Land-share coefficients are reported per 10 pp.
- X_i: log 1951 population, log 1950 cultivated area, share of owners with at least 3 ha, tenant-operated land share, log 1961 crop output value, and 1961 double-cropped paddy share.
- County and year fixed effects. Migration regressions also include birth-year fixed effects.

u indexes individual records where applicable, i historical townships, and t observation years. Previous estimation shows that tenant dryland shares are closer to redistributed dryland shares than total dryland shares.

Land reform and long-run economic outcomes

Outcome	Tenant dryland	Township dryland	Public-land share	Observations
Financial and business services	-0.107^{***} (0.029)	-0.079^{*} (0.038)	–	311,653
Education-industry employment	-0.144^{***} (0.044)	-0.068 (0.056)	–	474
Agriculture, forestry, fishing, and animal husbandry employment	–	–	2.353^{**} (0.842)	1,135
Agricultural-to-nonagricultural transition	-0.556^{***} (0.164)	-0.460^{**} (0.164)	-0.846^{**} (0.348)	30,886
Historical controls	Six	Six	Six	
County and year FE	Yes	Yes	Yes	

County-clustered SE in parentheses. Public-land agricultural employment pools all five censuses. Public-land transition also controls for township dryland and predicted private redistribution. Financial/business services include finance, insurance, real estate, and business services. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Township mayoral elections

Election sample

- Seven elections, 1998–2022
- 1,198 observations
- 232 historical townships in 15 counties

Outcomes

1. Number of officially listed candidates
2. Winner's share of valid candidate votes
3. Probability that a Kuomintang (KMT) candidate is elected, reported in percentage points

Township mayoral election outcomes

Outcome	Tenant dryland	Township dryland
Number of candidates	0.039^{**} (0.017)	0.050^{**} (0.019)
Winner's vote share (pp)	-0.671^{**} (0.258)	-0.672^{**} (0.266)
Probability of a KMT victory (pp)	1.819 (1.257)	2.300^{**} (0.994)
Historical controls	Six	Six
County and election-year FE	Yes	Yes
Observations	1,198	1,198
Historical townships	232	232

Separate regressions for each outcome and land measure. Land-share coefficients: per 10 pp. Candidate-count effects are in persons; vote-share and KMT-election effects are in percentage points. County-clustered SE in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

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- **Mobility:** Cash compensation made land wealth mobile and enabled local elites to leave. Dryland exposure increases migration; migrants hold more financial assets and have higher economic status and human capital.
- **Local economy:** Greater dryland exposure reduces local financial and business services, education employment, and agricultural-to-nonagricultural transitions.
- **Local politics:** Elite outflows disperse local political power and facilitate KMT entry. More dryland means more mayoral candidates, lower winners' vote shares, and a higher probability of KMT victory.
- **Public-land sales:** Attracting cultivators leaves townships more reliant on agriculture, with higher agricultural employment and fewer agricultural-to-nonagricultural transitions.

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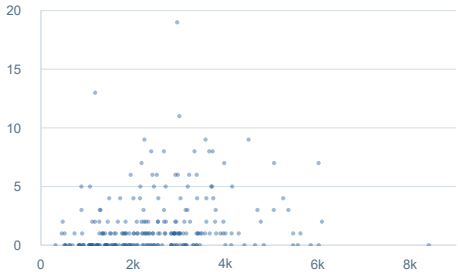
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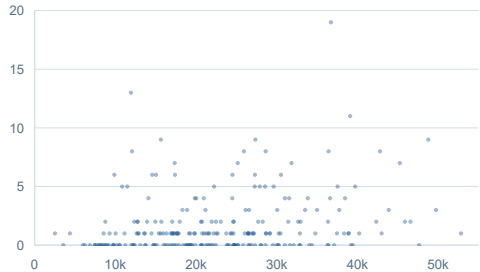
Large landowners and township size

1950 cultivated land (chia)



$r = 0.199$

1951 population (persons)



$r = 0.245$

Owner counts rise only modestly with township size. Population levels retain the direct interpretation in persons.

$N = 237$. Vertical axis: private-land owners with at least 30 hectares (about 30.9 chia). The 1950 survey reports hectares in bins; r denotes the Pearson correlation.

Step 1: observed inputs from the 1950 land survey

Private cultivated-land owners by holding-size bin

For township i, the survey reports n_{ib} owners in each of 12 ownership-area bins b:

$[0, 0.5)$, $[0.5, 1)$, $[1, 2)$, $[2, 3)$, $[3, 5)$, $[5, 7)$ ha
$[7, 10)$, $[10, 20)$, $[20, 30)$, $[30, 50)$, $[50, 100)$, $[100, \infty)$ ha

$$N_i = \sum_{b=1}^{12} n_{ib}$$

is the number of owners represented in the grouped distribution.

Other township-level fields

- total cultivated area C_i
- paddy area and dryland area Dry_i
- cultivated area by operating status, including tenant-operated paddy and dryland

Original source title: 《耕地所有權與經營狀況調查報告書》. The owner-size bins are measured in hectares; later calculations convert hectares and chia using $1 \text{ chia} = 0.969917$ hectares.

Step 2: fit a township owner-size distribution

Recommended distribution

$$F_i(a) = \Pr(A_i \le a) = \Phi\left(\frac{\ln a - \mu_i}{\sigma_i}\right)$$

County-bin shares $\pi_{c(i)b}$ supply limited shrinkage for small townships:

$$\widetilde{n}_{ib} = n_{ib} + \kappa \pi_{c(i)b}, \qquad \kappa = 25.$$

Parameters are estimated from the grouped likelihood:

$$(\widehat{\mu}_i, \widehat{\sigma}_i) = \arg\max_{\mu,\sigma} \sum_b \widetilde{n}_{ib} \ln[F_i(u_b) - F_i(l_b)] \,.$$

What the shrinkage does

- Adds 25 pseudo-owners distributed according to the 1950 county bin shares.
- Stabilizes the bin proportions of townships with few observed owners.
- Uses the 1950 land survey; N_i and fitted excess area determine each township estimate.

Model comparison

Five candidates are compared: lognormal, log-logistic, gamma, generalized gamma, and lognormal with a Pareto tail. The selected specification is lognormal with $\kappa = 25$.

The township likelihood uses the 1950 survey inputs. The 1953 county totals enter model comparison and validation.

Step 3: construct treatment intensity

Expected land above a uniform threshold

Let $h = 3$ chia (2.909751 hectares). For one owner,

$$e_i = \mathbb{E}[(A_i - h)_+] = \int_h^\infty [1 - F_i(a)]\, da.$$

For a lognormal distribution,

$$e_i = \exp\left(\widehat{\mu}_i + \frac{\widehat{\sigma}_i^2}{2}\right) \Phi\left(\frac{\widehat{\mu}_i + \widehat{\sigma}_i^2 - \ln h}{\widehat{\sigma}_i}\right) - h\Phi\left(\frac{\widehat{\mu}_i - \ln h}{\widehat{\sigma}_i}\right).$$

Predicted township redistribution is

$$\widehat{R}_i = N_i e_i,$$

after converting the result back from hectares to chia.

Dryland component and normalization

$$q_i = \frac{Dry_i^{\text{tenant}}}{C_i^{\text{tenant}}}, \qquad \widehat{R}_i^D = \widehat{R}_i q_i,$$

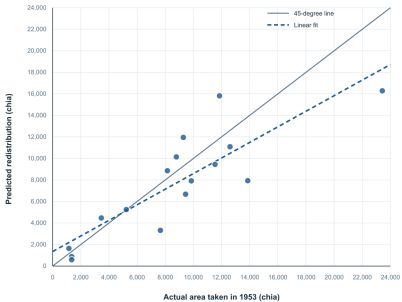
$$D_i = \frac{\widehat{R}_i^D}{C_i}.$$

Assume redistributed land has the same dryland share as tenant-operated cultivated land in 1950.

D_i is the regression treatment: predicted redistributed dryland divided by township cultivated land.

Validation against 1953 county totals

Construct	Predicted area	Actual area	Predicted / actual	Level corr.	Log corr.
Total redistributed land	122,313	139,003	0.880	0.852	0.923
Dryland: tenant land mix	30,204	21,385	1.412	0.800	0.886



Long-gap demographic DID: 1916–19 to 1966

Outcome (annual rate per 1,000)	DID / 10 pp	County SE	Wild p
Population change	-1.091	2.668	0.789
Birth rate	$+0.542$	0.530	0.348
Death rate	-0.124	0.565	0.852
Natural increase rate	$+0.665$	0.971	0.598
Implied migration / registration residual	-1.757	2.710	0.661

- 230 historical townships in 16 counties; 1916–19 annual rates versus 1966 vital rates.
- Treatment: predicted redistributed dryland / all cultivated land; county and scale controls.
- The pre-period joint test for natural increase rejects parallel trends ($p = 0.004$).

Post-period population change is an annualized 1955–66 rate, while birth and death rates are for 1966 alone. The residual is implied rather than observed migration. This long-gap comparison is exploratory.

Financial and business services: township-year comparisons

	Workplace Tenant dryland	Workplace Township dryland	Residence Tenant dryland	Residence Township dryland
Tenant dryland share	-0.080^{***} (0.015)	–	-0.066^{**} (0.029)	–
Township dryland share	–	-0.050^{***} (0.016)	–	-0.034 (0.029)
Years	1978–2000	1978–2000	1980–1989, 1992	1980–1989, 1992
Historical controls	Six	Six	Six	Six
County and year FE	Yes	Yes	Yes	Yes
Observations	3,823	3,823	1,756	1,756
Historical townships	232	232	228	228
County clusters	15	15	15	15
R^2	0.213	0.211	0.119	0.117

Land-share coefficients: per 10 percentage points. Share outcomes: percentage points. Parentheses: historical-county CR1 clustered standard errors. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$; two-sided $t(G-1)$ tests.

Source: selected regression tables, October 24, 2026, Table 5.

Equal township-year weights. Respondents aged 25–64; at least 20 unweighted observations per township-year. Workplace: Manpower Utilization Survey. Residence: Internal Migration Survey.

Employment outcomes: full coefficients

	Finance T	Finance D	Transition T	Transition D	Education T	Education D
	-0.107^{***}		-0.556^{***}		-0.144^{***}	
Tenant dryland share	(0.029)	–	(0.164)	–	(0.044)	–
		-0.079^{*}		-0.460^{**}		-0.068
Township dryland share	–	(0.038)	–	(0.164)	–	(0.056)
	1.904^{***}	1.950^{***}	0.949	1.208	1.156^{**}	1.254^{**}
Log population, 1951	(0.419)	(0.431)	(1.749)	(1.919)	(0.456)	(0.472)
	-0.546	-0.525	3.847^{**}	3.801^{**}	0.053	-0.082
Log cultivated area, 1950	(0.480)	(0.489)	(1.379)	(1.503)	(0.433)	(0.446)
	0.202	0.209	-0.229	-0.158	0.282	0.285
Owners with ≥ 3 ha (share)	(0.217)	(0.221)	(0.920)	(0.906)	(0.183)	(0.185)
	0.029	0.014	-0.112	-0.173	0.030	0.021
Tenant-operated land share	(0.069)	(0.063)	(0.395)	(0.400)	(0.112)	(0.123)
	-0.371	-0.390^{*}	-3.092	-3.136	-0.431^{**}	-0.327^{*}
Log crop output value, 1961	(0.213)	(0.197)	(2.157)	(2.138)	(0.173)	(0.175)
	0.142^{*}	0.156^{**}	0.447^{*}	0.524^{**}	0.068	0.076
Double cropping, 1961	(0.070)	(0.069)	(0.235)	(0.224)	(0.061)	(0.060)
County and year FE	Yes	Yes	Yes	Yes	Yes	Yes
Observations	311,653	311,653	30,886	30,886	474	474
Historical townships	232	232	232	232	237	237
County clusters	15	15	15	15	16	16
R^2	0.010	0.010	0.014	0.014	0.314	0.298

County-clustered SE in parentheses. Land shares: per 10 pp. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$. Source: Table 1. T/D: tenant/township dryland.

Migration: full coefficients

	Crop T	Crop D	+ Double T	+ Double D
	3.636^{**}		3.842^{**}	
Tenant dryland share	(1.626)	–	(1.487)	–
		4.513^{***}		5.336^{***}
Township dryland share	–	(1.450)	–	(1.633)
	12.283	12.273	11.567	11.696
Log population, 1951	(10.158)	(11.075)	(10.443)	(11.567)
	-3.819	-2.975	-7.777	-9.315
Log cultivated area, 1950	(7.038)	(5.912)	(8.618)	(8.255)
	1.184	0.691	0.404	-0.511
Owners with ≥ 3 ha (share)	(5.017)	(4.823)	(4.928)	(4.606)
	1.108	1.625	1.666	2.558
Tenant-operated land share	(2.195)	(2.466)	(1.730)	(1.907)
	-3.639	-3.443	1.814	5.073
Log crop output value, 1961	(5.649)	(5.127)	(6.603)	(5.366)
			-2.079	-2.965^{*}
Double cropping, 1961	–	–	(1.411)	(1.537)
County and year FE	Yes	Yes	Yes	Yes
Birth-year FE	Yes	Yes	Yes	Yes
Observations	692	692	692	692
Historical townships	200	200	200	200
County clusters	16	16	16	16
R^2	0.118	0.125	0.123	0.135

County-clustered SE in parentheses. Land shares: per 10 pp. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$. Source: Table 2. T/D: tenant/township dryland.

Election outcomes: full coefficients

	Candidates T	Candidates D	Vote share T	Vote share D	KMT T	KMT D
Tenant dryland share	0.039^{**}	–	-0.671^{**}	–	1.819	–
	(0.017)	0.050^{**}	(0.258)	-0.672^{**}	(1.257)	2.300^{**}
Township dryland share	–	(0.019)	–	(0.266)	–	(0.994)
	0.287^{**}	0.290^{***}	-2.528	-2.392	3.265	3.383
Log population, 1951	(0.099)	(0.093)	(1.684)	(1.543)	(5.884)	(5.594)
	-0.115	-0.137	-0.386	-0.374	-4.242	-5.203
Log cultivated area, 1950	(0.097)	(0.100)	(1.845)	(1.457)	(5.519)	(4.250)
	-0.019	-0.028	0.574	0.689	2.448	2.033
Owners with ≥ 3 ha (share)	(0.042)	(0.043)	(0.772)	(0.830)	(3.190)	(3.329)
	0.009	0.020	-0.886	-1.037	0.421	0.934
Tenant-operated land share	(0.032)	(0.030)	(0.598)	(0.636)	(1.326)	(1.281)
	0.130	0.151	-0.665	-0.733	1.987	2.930
Log crop output value, 1961	(0.099)	(0.108)	(0.777)	(0.813)	(6.347)	(5.358)
	0.024	0.017	-0.506^{**}	-0.407	0.787	0.466
Double cropping, 1961	(0.017)	(0.017)	(0.222)	(0.240)	(0.503)	(0.457)
County and year FE	Yes	Yes	Yes	Yes	Yes	Yes
Observations	1,198	1,198	1,198	1,198	1,198	1,198
Historical townships	232	232	232	232	232	232
County clusters	15	15	15	15	15	15
R^2	0.091	0.093	0.064	0.064	0.186	0.188

County-clustered SE in parentheses. Land shares: per 10 pp. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$. Source: Table 3. T/D: tenant/township dryland.

Public-land outcomes: full coefficients

	Ag. pooled	Ag. occupation	Ag. industry	Transition 1	Transition 2	Transition 3
						-0.303
Township dryland share	–	–	–	–	–	(0.262)
	2.353^{**}	2.012^{**}	2.999^{**}	-1.167^{***}	-1.065^{***}	-0.846^{**}
Leased public-land share	(0.842)	(0.764)	(1.244)	(0.189)	(0.196)	(0.348)
						0.168
Private redistribution	–	–	–	–	–	(0.406)
	-15.503^{***}	-16.523^{***}	-10.000^{***}	2.187	1.909	1.282
Log population, 1951	(1.837)	(1.837)	(2.170)	(1.746)	(1.870)	(2.559)
	4.747	6.625^{*}	-2.421	0.827	1.623	2.775
Log cultivated area, 1950	(3.291)	(3.261)	(3.399)	(1.658)	(1.533)	(2.352)
	-0.596	-1.125	1.240	-0.073	0.053	-0.252
Owners with ≥ 3 ha (share)	(1.233)	(1.212)	(1.632)	(1.034)	(0.963)	(1.002)
	-0.689	-0.558	-0.944	0.088	-0.049	-0.102
Tenant-operated land share	(0.668)	(0.605)	(0.814)	(0.341)	(0.390)	(0.415)
	3.610	3.197	4.368^{*}	-0.909	-1.710	-2.362
Log crop output value, 1961	(3.241)	(3.565)	(2.477)	(2.143)	(1.935)	(1.899)
	-1.282^{*}	-1.331^{*}	-0.958		0.423^{*}	0.455^{*}
Double cropping, 1961	(0.654)	(0.638)	(0.713)	–	(0.236)	(0.254)
County and year FE	Yes	Yes	Yes	Yes	Yes	Yes
Observations	1,135	908	454	30,886	30,886	30,886
Historical townships	227	227	227	232	232	232
County clusters	16	16	16	15	15	15
R^2	0.718	0.731	0.525	0.014	0.014	0.014

County-clustered SE in parentheses. Land shares: per 10 pp. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$. Source: Table 4. T/D: tenant/township dryland.

Township-year services: full coefficients

	Workplace T	Workplace D	Residence T	Residence D
	-0.080^{***}		-0.066^{**}	
Tenant dryland share	(0.015)	–	(0.029)	–
		-0.050^{***}		-0.034
Township dryland share	–	(0.016)	–	(0.029)
	1.594^{***}	1.636^{***}	0.882^{***}	0.920^{***}
Log population, 1951	(0.370)	(0.380)	(0.182)	(0.181)
	-0.759^{**}	-0.800^{***}	-0.359	-0.405^{**}
Log cultivated area, 1950	(0.268)	(0.265)	(0.211)	(0.185)
	0.332^{**}	0.337^{**}	0.087	0.087
Owners with ≥ 3 ha (share)	(0.124)	(0.127)	(0.215)	(0.220)
	-0.020	-0.027	-0.051	-0.054
Tenant-operated land share	(0.032)	(0.033)	(0.049)	(0.050)
	-0.047	-0.020	0.021	0.054
Log crop output value, 1961	(0.293)	(0.293)	(0.174)	(0.193)
	0.113^{**}	0.119^{**}	0.104	0.107
Double cropping, 1961	(0.044)	(0.043)	(0.065)	(0.061)
County and year FE	Yes	Yes	Yes	Yes
Observations	3,823	3,823	1,756	1,756
Historical townships	232	232	228	228
County clusters	15	15	15	15
R^2	0.213	0.211	0.119	0.117

County-clustered SE in parentheses. Land shares: per 10 pp. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$. Source: Table 5. T/D: tenant/township dryland.